

Single-Phase Machine

A single-phase induction motor is an electric motor that operates on a single-phase AC power supply and is widely used in domestic and light commercial applications due to its simple, robust, and cost-effective design.

Construction

A single-phase induction motor consists of two main parts: a stationary stator and a rotating rotor.

Stator: This is the stationary part, made of laminated steel to reduce energy losses. It contains the main winding and, in most types, an auxiliary (starting) winding, displaced by 90 electrical degrees from the main winding.

Rotor: The rotor is typically a squirrel-cage design, consisting of conductive bars (made of aluminum or copper) embedded in a laminated core and short-circuited by end rings. The rotor bars are often skewed to help prevent magnetic locking and ensure quieter, smoother operation.

Other components: Bearings support the rotor shaft, and some motors include a centrifugal switch or capacitor as part of the starting mechanism.



Working Principle

When a single-phase AC supply is applied to the stator's main winding, it produces a pulsating magnetic field, not a naturally rotating one. According to the double revolving field theory, this pulsating field can be seen as two equal and oppositely rotating magnetic fields, which cancel each other out at a standstill, resulting in zero net starting torque.

To **become self-starting**, the motor must employ an additional mechanism to create a phase difference in the magnetic field, simulating a rotating magnetic field during startup. This is commonly achieved using an **auxiliary winding** in conjunction with a capacitor or high resistance, which causes the current in the

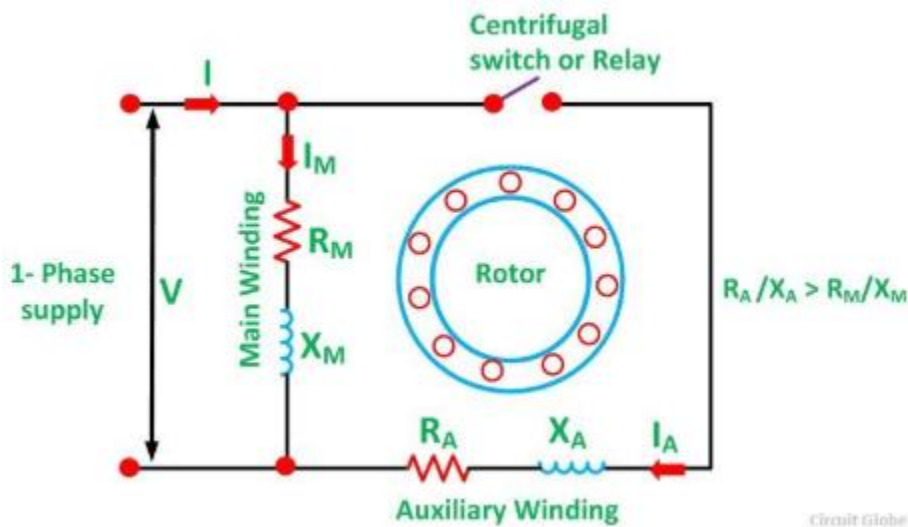
auxiliary winding to be out of phase with the current in the main winding. Once the motor reaches a sufficient speed (typically 75–80% of synchronous speed), a centrifugal switch often disconnects the auxiliary winding and starting components. The motor then continues to run on the main winding alone, relying on the induced current in the rotor to generate the necessary torque.



Types of Single-Phase Induction Motors

Single-phase induction motors are classified based on their starting methods:

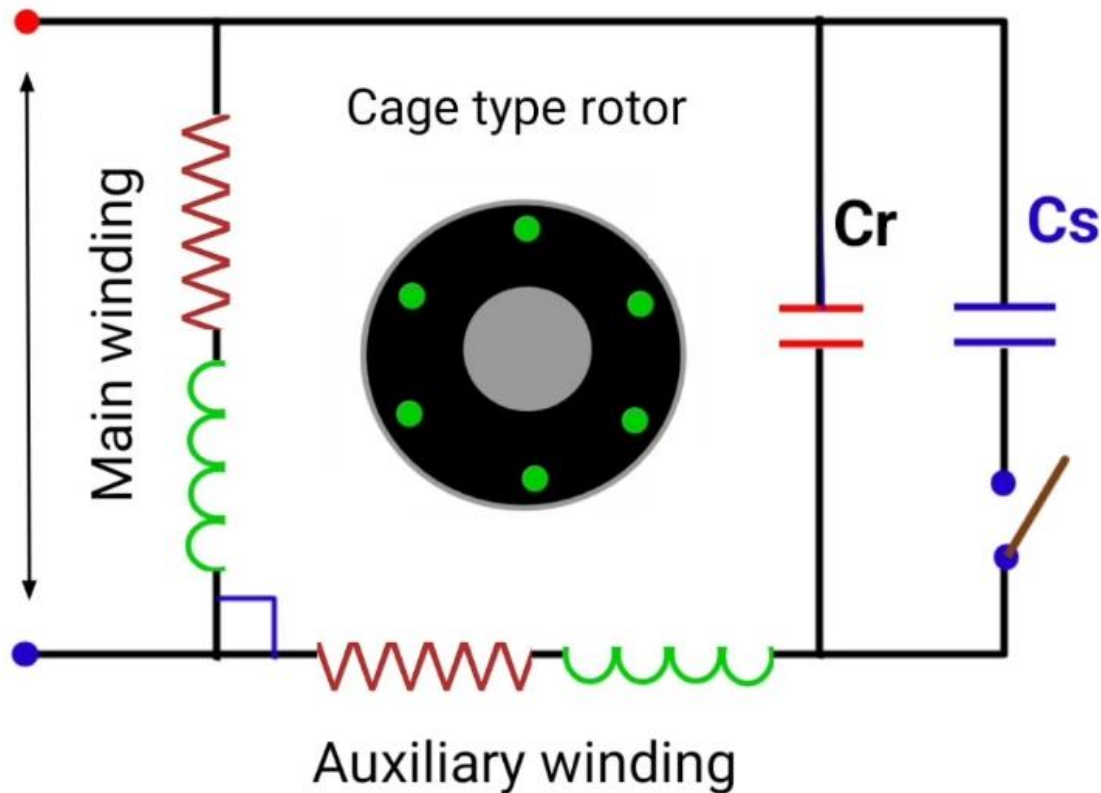
Split-Phase (Resistance Start) Motor: Uses a high-resistance auxiliary winding to create a small phase difference. They have low starting torque and are used in fans and small pumps.



Capacitor-Start Motor: Adds a capacitor in series with the auxiliary winding to produce a larger phase shift, resulting in high starting torque, making them suitable for compressors and large pumps.

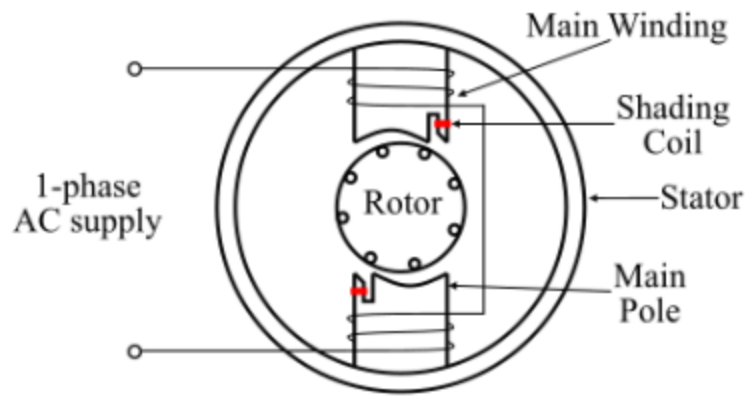
Capacitor-Start Capacitor-Run Motor: Uses two capacitors (one for starting, one for running) to offer both high starting torque and

improved running efficiency and power factor. They are used in air conditioners and high-duty blowers.



Permanent-Split Capacitor (PSC) Motor: Features a single, permanently connected capacitor, eliminating the need for a centrifugal switch. This provides smooth, quiet operation and moderate starting torque for continuous-duty applications like ceiling fans.

Shaded-Pole Motor: The simplest and least expensive type, using a short-circuited copper band (shading coil) on the poles to create a weak rotating field. They have very low starting torque and are used in small fans and toys.



Applications

Single-phase induction motors are widely used in residential, commercial, and light industrial settings where a three-phase power supply is unavailable.

Household Appliances: Fans (ceiling, table, exhaust), refrigerators, washing machines, vacuum cleaners, mixers, and hair dryers.

Commercial & Industrial: Water pumps, air conditioners, small air compressors, blowers, and various machine tools like drilling machines and lathes.

Advantages and Disadvantages

Advantages:

- Simple and robust construction, resulting in lower cost and easier maintenance.
- Compatible with standard single-phase power supplies widely available in homes and offices.

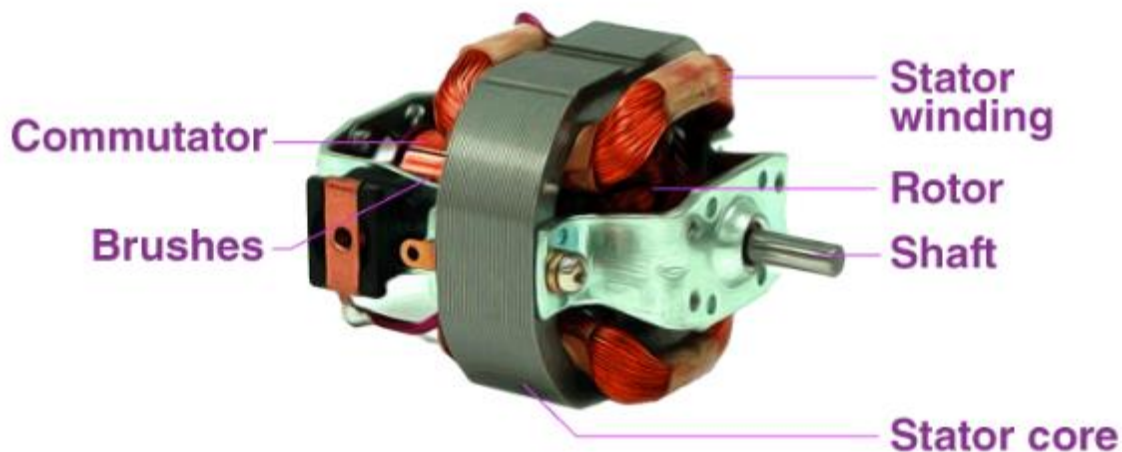
Disadvantages:

- Not self-starting (requires auxiliary mechanisms).
- Lower efficiency and power factor compared to three-phase motors.
- Lower starting torque and limited power range, generally available in fractional-kilowatt sizes.

Universal Motor

Single-phase AC series motor is a Universal Motor, characterized by its ability to run on both AC and DC power due to series-connected field and armature windings, offering high speed and torque for tools, vacuums, and mixers.

Lab testing involves measuring performance (speed, torque, power, efficiency) under load using dynamometers, checking winding health (multimeter for shorts), and analyzing characteristics like no-load speed (which can be very high) to understand efficiency, power factor, and thermal limits, often using test benches with AC/DC supplies and measurement systems.



Key Characteristics & Working Principle

- Series Connection: Field windings are in series with the armature, similar to a DC series motor.

- **AC/DC Versatility:** Reversing current in both field and armature simultaneously results in unidirectional torque, allowing operation on AC or DC.
- **High Speed, High Torque:** Provides high starting torque and operates at high speeds, ideal for portable appliances.
- **Commutation:** Uses brushes and a commutator, requiring careful brush placement for optimal AC performance.
- **Construction:** Stator/yoke is laminated to reduce eddy current losses, and some have compensating windings for better AC performance.

Single-Phase Motor Types Summary:

1. **Repulsion Motor** – high starting torque, brush wear issues.
2. **Single-Phase Synchronous Motor** – constant speed, complex starting.
3. **Brushed DC Motor** – simple, low cost, brush wear.
4. **BLDC Motor** – high efficiency, quiet, expensive.

1. **Repulsion Motor (Single-Phase AC) Principle:** Uses magnetic repulsion between stator & rotor fields; rotor is short-circuited via brushes on a commutator, allowing induced currents to create torque. **Key Feature:** High starting torque, good for heavy loads. **Drawback:** Brush/commutator wear, sparking, poor power factor at low speeds. **Uses:** Older lifts, pumps, hoists, compressors.

2. Single-Phase Synchronous Motor Principle: Rotor locks into synchronous speed with the rotating magnetic field of the stator, often needing a starting mechanism (like a shaded pole or separate winding). Key Feature: Constant speed operation, precise timing. Drawback: Can be complex to start, requires DC excitation for rotor field in some designs.

3. Brushed DC Motor (DC Powered) Principle: Current flows to the rotor (armature) via brushes and a commutator, reversing current direction to maintain rotation against permanent magnet or field windings. Key Feature: Simple, direct DC power, low cost. Drawback: Brush wear, arcing, shorter lifespan, noise. Uses: Toy cars, printers, battery-operated devices.

4. Brushless DC (BLDC) Motor (Often Single-Phase Driven) Principle: Replaces mechanical commutation with electronic switching (transistors) to energize stator windings, with permanent magnets on the rotor. Key Feature: High efficiency, longer life, quiet, high speed/acceleration, good for cooling fans, drives. Drawback: Higher cost due to required electronics (driver/controller)